Air University

Student ID: 242242

AIR UNIVERSITY

Final Term Examinations: Spring 2025

~~Subjective Part~~

(To be solved on Answer Books only)

Subject: Introduction to Software Engineering
Class: BSCYS-II
Section(s): A & B
Course Code: SE101

Time Allowed: 180 Minutes
Max Marks: 100
FM's Name: Muhammad Jalal Shah
FM's Signature:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Tables / calculators are allowed / not allowed.

	Q. No. 1 (CLO 1)	30 Marks
a	A software company released an urgent hotfix for their customer relationship management (CRM) system. Hotfix addressed a display issue where the "Notes" section of a customer's profile was not rendered correctly on certain browsers. After deploying hotfix, a QA engineer checks only one specific customer profile, navigates to the "Notes" section, and verifies that the content is now visible and formatted correctly on the affected browser. They do not check other sections of the CRM or other customer profiles. **Question:** Based on the scenario provided, what specific category of software testing is being conducted? Justify your answer.	10
b	Explain the core reasons for performing stress testing, including the critical objectives it aims to achieve and the specific types of system behaviors or vulnerabilities it is designed to reveal.	10
c	Is software testing considered a part of Quality Assurance (QA), Quality Control (QC), or both? Explain your reasoning.	10
	Q. No. 2 (CLO 4)	25 Marks
	Scenario: University Final Year Project Portal (FYP-Portal) Academica University" recognizes the growing importance of practical, project-based learning, especially for its final-year students. They currently struggle with inconsistent project supervision, difficulty in tracking student progress, and a lack of organized access to past project work for future students. To address these	

challenges, the university plans to develop a new **Final Year Project Portal (FYP-Portal)**.

This portal aims to centralize the management, submission, tracking, and archiving of final year projects, providing significant benefits for students, faculty, and the institution.

The FYP-Portal will facilitate:

Core Functionalities:

- **Project Proposal & Approval:** Students submit project proposals; faculty supervisors and department heads review and approve them.
- **Progress Tracking:** Students log their project progress (milestones, hours worked, challenges); supervisors monitor progress and provide feedback.
- **Document Submission & Version Control:** Students upload project deliverables (reports, code, presentations); the system manages versions.
- **Evaluation Management:** Faculty members submit project evaluations and grades.
- **Project Repository:** Approved and completed projects are securely stored and indexed, creating a searchable knowledge base for future students and faculty.
- **Notification System:** Automated alerts for deadlines, approvals, feedback, etc.

External Entities:

- **Final Year Students:** Interact with the system to submit proposals, log progress, upload deliverables, view supervisor feedback, and view their final grades.
- **Project Supervisors (Faculty):** Use the system to review proposals, approve projects, provide feedback on progress, access student submissions, and submit evaluations/grades.
- **Department Heads/Coordinators (Faculty):** Review and approve proposals, assign supervisors, monitor overall project progress within their department, and access final project archives.
- **University Administration/Registrar:** Access final project grades for academic record keeping and potentially retrieve project data for accreditation or reporting.
- **New/Junior Students:** Access the public-facing project repository to browse and search completed projects for inspiration and reference.

Key Data Flows:

- **Project Proposals:** Flow from Final Year Students to the FYP-Portal, then to Project Supervisors and Department Heads for review.
- **Approval/Rejection Decisions:** Flow from Project Supervisors/Department Heads to the FYP-Portal, and then to Final Year Students.

a

- **Progress Updates:** Flow from Final Year Students to the FYP-Portal, then to Project Supervisors.
- **Feedback/Comments:** Flow from Project Supervisors to the FYP-Portal, then to Final Year Students.
- **Project Deliverables (Reports, Code):** Flow from Final Year Students to the FYP-Portal.
- **Evaluation Forms/Grades:** Flow from Project Supervisors to the FYP-Portal.
- **Final Project Data:** Flows from FYP-Portal to University Administration/Registrar.
- **Search Queries:** Flow from New/Junior Students to the FYP-Portal (repository).
- **Search Results/Archived Projects:** Flow from FYP-Portal (repository) to New/Junior Students.
- **Notification Alerts:** Flow from FYP-Portal to Final Year Students, Project Supervisors, and Department Heads.

Questions:

1. Draw a Context Diagram for the University Final Year Project Portal (FYP-Portal).
2. Develop a Level 1 Data Flow Diagram (DFD) for the University Final Year Project Portal (FYP-Portal).

Q. No. 3 (CLO 1) — **25 Marks**

Scenario: Launching a New Software Feature (Project "Aurora")

Your team is tasked with launching "Aurora," a significant new feature for an existing software product. The project manager has broken down the effort into the following key activities:

Activity	Predecessor	Duration (Days)
A (Market Research)	---	5
B (Design UI/UX)	A	4
C (Backend Dev)	A	6
D (Frontend Dev)	B	5
E (Database Setup)	C	3
F (Integration)	D, E	4
G (Testing)	F	3
H (Documentation)	F	2
I (Deployment)	G, H	1

	Questions: 1. **Draw a network diagram** representing the project activities and their dependencies. 2. **Calculate the float (slack)** for each activity. Float refers to the amount of time an activity can be delayed without impacting the overall project completion time. 3. **Identify the critical path**, which is the longest sequence of dependent activities that determines the minimum project duration. 4. **Determine the total project duration**, assuming there are no delays.	
	Q. No. 4 (CLO 3)	**20 Marks**
a	Apply your knowledge of Object-Oriented Design (OOD) principles to explain how they can contribute to creating scalable software solutions. Furthermore, discuss how the constraints of scope, time, and cost influence the quality of the final software product.	5+5
b	An inertial navigation system is a device that uses gyros and accelerometers to measure changes in its position and orientation. It is typically used in aircraft to provide a backup navigation mechanism for GPS and other radio-based systems and to provide altitude reference when flying on autopilot. You have been hired to develop such a device to be the main navigation system for an unmanned submarine. Describe the development methodology you would use, with particular reference to how you would assure your customer of the device's dependability and what sort of problems you would watch out for.	10

***************** End of Question Paper *****************